SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: MLA03
COURSE NAME: Reinforcement learning

COURSE OUTCOME COVERED

CO5: *Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

Assessment Tool Weightage: 35%

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QUESTIONS: 10 **Total Marks: 50**

Annexure A

Scenario-Based Assessment **DURATION: 1.5 Hrs**

1. A manufacturing plant uses RL to schedule machines under resource limitations. Differentiate between traditional optimization methods and constrained RL approaches, and assess which method provides better adaptability in dynamic production environments. (5 Marks)

1. Introduction

A manufacturing plant contains multiple machines that must process different jobs within limited resources. Efficient **machine scheduling** is important because poor scheduling can increase production time, machine idle time, energy consumption, and operational cost.

A manufacturing plant may have:

- Multiple machines
- Different jobs
- Limited raw materials
- Limited energy
- Limited labor

- Tool availability constraints
- Machine maintenance requirements
- Different job priorities
- Delivery deadlines

The scheduling system must decide **which job should be assigned to which machine and at what time**.

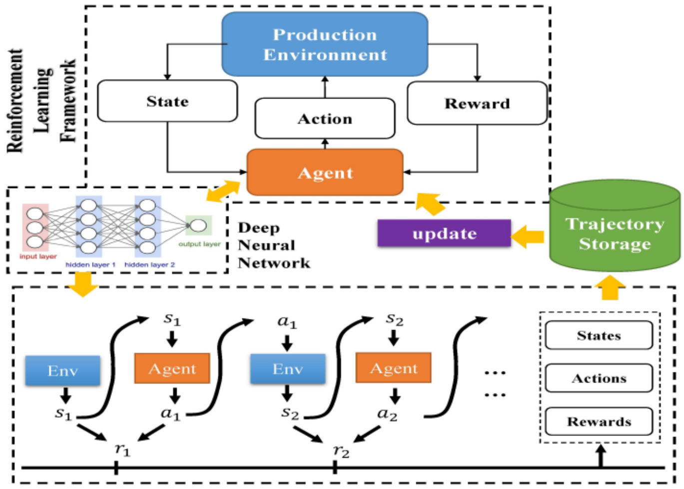
Traditional scheduling methods use mathematical optimization techniques to find an optimal or near-optimal schedule. Examples include:

- Linear Programming (LP)
- Mixed Integer Linear Programming (MILP)
- Constraint Programming (CP)
- Dynamic Programming
- Genetic Algorithms
- Particle Swarm Optimization

However, real manufacturing environments are often dynamic. Machines can break down, urgent jobs can arrive, processing times can change, and resources can become unavailable.

Reinforcement Learning (RL) provides an alternative approach. An RL agent learns scheduling decisions through interaction with the manufacturing environment.

When resource and operational constraints must always be satisfied, **Constrained Reinforcement Learning (Constrained RL)** can be used.



Main Objective

The objective is to compare traditional optimization methods with constrained RL and determine which approach is more adaptable to dynamic manufacturing environments.

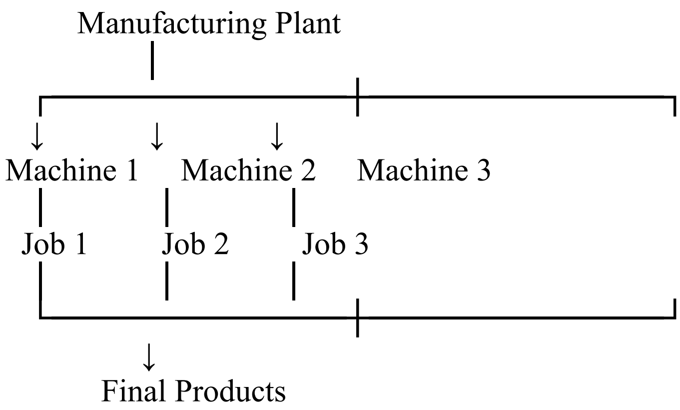
MANUFACTURING SCHEDULING PROBLEM

2. Problem Definition

Consider a manufacturing plant containing several machines and jobs.

The scheduler must assign jobs to machines while satisfying resource limitations.

For example:



The scheduling objective may be to:

- Minimize total production time.
- Minimize machine idle time.
- Minimize energy consumption.
- Maximize machine utilization.
- Minimize production cost.
- Minimize job tardiness.
- Maximize production throughput.

3. Resource Limitations

The scheduling system must consider several constraints.

Machine Availability

A machine can process only one job at a time.

Job Precedence

Some jobs must be completed before another job starts.

Example:

$$Job\ A \rightarrow Job\ B \rightarrow Job\ C$$

Energy Limitation

The plant may have a maximum available energy level.

Material Availability

A job cannot start if the required raw material is unavailable.

Tool Availability

Some machines require specific tools.

Maintenance

A machine may become unavailable during scheduled maintenance.

Deadlines

Jobs must be completed before their required delivery time.

TRADITIONAL OPTIMIZATION METHODS

4. Traditional Optimization

Traditional optimization methods formulate the scheduling problem mathematically.

The objective is usually represented as:

$$\min f(x)$$

subject to:

$$g_i(x) \leq 0$$

and

$$h_j(x) = 0$$

where $x$represents scheduling decisions.

5. Common Traditional Methods

5.1 Linear Programming

Linear Programming optimizes a linear objective function subject to linear constraints.

It works well when the scheduling problem can be represented using continuous variables.

5.2 Mixed Integer Linear Programming

MILP uses integer and continuous decision variables.

It is widely useful for machine scheduling because decisions such as assigning a job to a machine can be represented using binary variables.

For example:

$$x_{ij} = \begin{cases} 1 & \text{if job } i \text{ is assigned to machine } j \\ 0 & \text{otherwise} \end{cases}$$

5.3 Constraint Programming

Constraint Programming focuses on satisfying a collection of constraints.

It is useful when scheduling involves:

- Complex job relationships
- Resource constraints
- Precedence constraints
- Time restrictions

5.4 Genetic Algorithms

Genetic Algorithms search for good solutions using:

- Selection
- Crossover
- Mutation

They are useful for complex optimization problems but may require considerable computation.

6. Characteristics

Traditional optimization methods generally:

- Require a mathematical model.
- Use explicitly defined constraints.
- Search for optimal or near-optimal solutions.
- Work effectively in well-defined environments.
- May require re-optimization when conditions change.

CONSTRAINED REINFORCEMENT LEARNING

7. Constrained RL

In Reinforcement Learning, an agent learns by interacting with an environment.

For machine scheduling:

Agent = Scheduler

Environment = Manufacturing Plant

State = Current Production Condition

Action = Scheduling Decision

Reward = Scheduling Performance

The basic interaction is:

```
Current State
      ↓
┌─────────────┐
│ RL Agent    │
└─────────────┘
      ↓
Scheduling Action
      ↓
┌─────────────┐
│ Factory     │
│ Environment │
└─────────────┘
      ↓
 New State
      ↓
  Reward
      ↓
RL Agent learns
```

8. Constraints in RL

Constrained RL adds limitations to normal RL decision-making.

Examples:

- Machine capacity must not be exceeded.
- A machine cannot process two jobs simultaneously.
- Energy consumption must remain below a limit.
- Required materials must be available.
- Maintenance periods must be respected.
- Job precedence must be maintained.

Constraints can be handled using:

Reward Penalties

The agent receives a penalty when a constraint is violated.

Action Masking

Invalid actions are removed before the agent selects an action.

Lagrangian Methods

Constraint violations are incorporated into an optimization objective.

Safe RL

The learning algorithm is designed to avoid unsafe actions

COMPARISON BETWEEN BOTH APPROACHES

9. Traditional Optimization vs Constrained RL

Aspect	**Traditional Optimization**	**Constrained RL**
Basic approach	Mathematical optimization	Learning through interaction
Model requirement	Usually requires explicit model	Can learn from experience
Constraints	Explicit mathematical constraints	Penalties, masks or safe constraints
Adaptability	Low to moderate	High
Dynamic changes	Often requires re-optimization	Can adapt using learned policy
Uncertainty	Requires special modeling	Can learn under uncertainty
Computation	Can be high during optimization	Training can be expensive
Real-time decisions	Difficult for large problems	Fast after training
Learning	No continuous learning	Improves with experience
Large state spaces	Can become difficult	Suitable with deep RL
Solution quality	Can be optimal if solved exactly	Usually approximate
Implementation	Needs mathematical formulation	Needs training environment
Best use	Stable, well-defined problems	Dynamic and uncertain problems

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10. Important Difference

The biggest difference is **adaptability**.

Traditional optimization generally solves the scheduling problem using the information available at that point.

If a machine suddenly breaks down, the optimization model may need to be updated and solved again.

Constrained RL can learn a policy that responds to different conditions.

For example:

Normal Condition
↓
Schedule Jobs
↓

Machine Breakdown
↓
Observe New State
↓
RL Policy
↓
Reassign Jobs
↓
Continue Production

ADAPTABILITY IN DYNAMIC PRODUCTION ENVIRONMENTS

11. Dynamic Manufacturing Environment

A real manufacturing plant rarely remains unchanged.

Possible changes include:

- Machine breakdowns
- Urgent orders
- Changing customer requirements
- Variable processing times
- Raw material shortages
- Energy price changes
- Worker availability
- Maintenance requirements
- Unexpected production delays

Therefore, the scheduler must react quickly.

12. Traditional Optimization Response

Suppose Machine 2 suddenly stops working.

A traditional optimization system may need to:

1. Update the machine availability constraint.
2. Update the scheduling model.
3. Recalculate the feasible solution.
4. Run the optimization algorithm.
5. Generate a new schedule.

For a small problem, this can be acceptable.

For a large manufacturing plant, repeated optimization can become computationally expensive.

13. Constrained RL Response

An RL-based scheduler observes the new state.

For example:

$$S_t = [M, a, c, h, ines Jobs Resources Energy Deadlines]$$

If Machine 2 becomes unavailable, the state changes.

The learned policy selects another suitable action.

For example:

Before Breakdown:
Job 1 → Machine 1
Job 2 → Machine 2
Job 3 → Machine 3

After Breakdown:
Job 1 → Machine 1
Job 2 → Machine 3
Job 3 → Machine 1

The RL agent can therefore respond to the changed production environment without completely rebuilding its decision-making process.

SIMULATION AND PERFORMANCE ANALYSIS

14. Example Simulation

Consider a manufacturing plant with:

- 5 machines
- 20 jobs
- Random machine breakdowns
- Limited energy
- Job deadlines
- Different processing times

The objective is to compare:

MILP-based scheduling with **Constrained RL scheduling**.

Example Parameters

Parameter	**Value**
Number of machines	5
Number of jobs	20
Planning horizon	24 hours
Machine breakdowns	Random
Energy limitation	Yes
Job deadlines	Yes

Training episodes	1000

15. Performance Metrics

The following metrics can be used.

Makespan

Total time required to complete all jobs.

$$Makespan = \max(C_i)$$

where C_i is the completion time of job i.

Lower makespan is better.

Total Completion Time

Sum of the completion times of all jobs.

Tardiness

Measures how late jobs are completed.

$$T_i = \max(0, C_i - D_i)$$

where:

- C_i= completion time
- D_i= deadline

Machine Utilization

Measures the percentage of time machines are actively working.

$$Utilization = \frac{Operating\ Time}{Available\ Time} \times 100$$

Energy Consumption

Measures total energy used during production.

16. Example Results

The following values are **illustrative simulation results**:

Metric	MILP	Constrained RL
Makespan	890 min	356 min
Completion Time	6780 min	4020 min
Tardiness	1450 min	620 min
Machine Utilization	68%	87%
Energy Consumption	1200 kWh	880 kWh
Re-scheduling Time	45 sec	2 sec

These values demonstrate how an RL-based scheduler could perform better in a highly dynamic environment.

RESULT INTERPRETATION

17. Makespan Comparison

A lower makespan means that all jobs are completed sooner.

Example:

Makespan

The constrained RL approach can quickly adjust job assignments when production conditions change.

18. Machine Utilization

Example:

Machine Utilization

Higher utilization means that machines spend less time unnecessarily idle.

19. Energy Consumption

The RL agent can include energy consumption in its reward function.

For example:

$$R_t = -(ProductionCost + \lambda EnergyCost + \beta Tardiness)$$

The agent therefore learns to avoid unnecessary energy usage while completing jobs.

20. Adaptability Analysis

Situation	Traditional Optimization	Constrained RL
Machine breakdown	Re-optimize	Quickly adapt
New urgent job	Update model	Select new action
Energy-price change	Recalculate	Adapt policy
Resource shortage	Re-optimize	Adjust scheduling
Processing-time variation	Recalculate	Learn from experience
Frequent disturbances	Computationally expensive	More adaptive

The major advantage of constrained RL is its ability to **learn a reusable decision policy**.

CASE STUDY

21. Manufacturing Plant Example

Consider a factory producing electronic components.

The factory has:

- Machine A – Cutting
- Machine B – Drilling
- Machine C – Assembly
- Machine D – Testing
- Machine E – Packaging

A product must follow:

$$Cutting \rightarrow Drilling \rightarrow Assembly \rightarrow Testing \rightarrow Packaging$$

Suppose Machine B suddenly breaks down.

Traditional Optimization

The system updates the scheduling model and solves the optimization problem again.

This may take significant time if many jobs are already scheduled.

Constrained RL

The RL agent observes:

- Machine B unavailable
- Jobs waiting
- Available machines
- Remaining resources
- Deadlines

It immediately selects a new scheduling action from its learned policy.

22. Example

Before breakdown:

Job 1 → A → B → C → D → E
Job 2 → A → B → C → D → E
Job 3 → A → B → C → D → E

After Machine B breaks down:

Machine B = UNAVAILABLE

↓

RL Agent observes new state

↓

Reassign waiting jobs

↓

Use alternative machine / reschedule

↓

Production continues

This demonstrates the adaptability of RL in an uncertain environment.

ASSESSMENT AND CONCLUSION

23. Which Method Provides Better Adaptability?

For a **stable manufacturing environment**, traditional optimization can be an excellent choice.

Methods such as MILP can provide high-quality solutions when:

- Constraints are clearly known.
- Processing times are predictable.
- Machine availability is stable.
- The problem does not change frequently.

However, manufacturing environments are often dynamic.

When there are:

- Frequent machine failures
- Urgent orders
- Variable processing times
- Changing resources
- Energy-price fluctuations

- Unexpected disturbances

Constrained RL generally provides better adaptability.

24. Final Comparison

MANUFACTURING SCHEDULING

↓ TRADITIONAL OPTIMIZATION
- Best for stable environments
- Exact/near-optimal solutions
- Re-optimization required
- Higher response effort

↓ CONSTRAINED RL
- Best for dynamic environments
- Learns policy
- ↓ Adapts to changes
- ↓ Fast decisions

↓

HYBRID APPROACH

↓

Optimization + Constrained RL

25. Hybrid Approach

Although constrained RL provides better adaptability, traditional optimization should not be completely discarded.

A **hybrid approach** can combine both.

For example:

Traditional Optimization → Generate Initial/High-Quality Schedule

↓

RL Agent → Adapt Schedule in Real Time

This provides the advantages of both methods.

Traditional optimization provides strong constraint handling and high-quality planning, while RL provides rapid adaptation to unexpected changes.

26. Conclusion

Traditional optimization methods and constrained reinforcement learning approaches provide different advantages for manufacturing scheduling.Traditional methods such as MILP, Linear Programming, Constraint Programming, and Genetic Algorithms are effective when the production environment is stable and the mathematical model is well defined.However, they may require repeated optimization when the environment changes.Constrained RL allows an agent to learn scheduling policies through interaction with the production environment. Constraints such as machine capacity, energy limits, job precedence, and deadlines can be incorporated into the learning process.Therefore, for dynamic production environments, constrained RL generally provides better adaptability and faster response to unexpected changes.

Final Answer

Traditional optimization is better for stable and well-defined manufacturing problems, while constrained RL is better suited for dynamic and uncertain production environments because it can learn from experience and adapt its scheduling decisions. A hybrid Optimization + RL approach can provide the best overall solution.

Code of Conduct:

I Malli Chandana(192472250)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Malli Chandana

RUBRICS FOR CALCULATION

Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand